



EE345 – 第三代半导体基础导论

Introduction to Wide-bandgap Semiconductors

Chapter 05

SiC devices and applications



Review of last chapter

- Characterization techniques for semiconductor material and devices:
 - Optical microscopy with Nomarski interference contrast
 - KOH etching
 - Scanning Electron Microscopy (SEM)
 - Cathodoluminescence (CL)
 - Hall Effect measurements
 - Atomic Force Microscopy (AFM)
 - Transmission Electron Microscope (TEM)
 - X-ray diffraction (XRD)
 - Secondary ion mass spectrometry (SIMS)
 - X-ray photoelectron spectroscopy (XPS)
 - Mercury Probe C–V
 - Photoluminescence (PL)
 - Raman spectroscopy
 - Ellipsometry
 - More ...

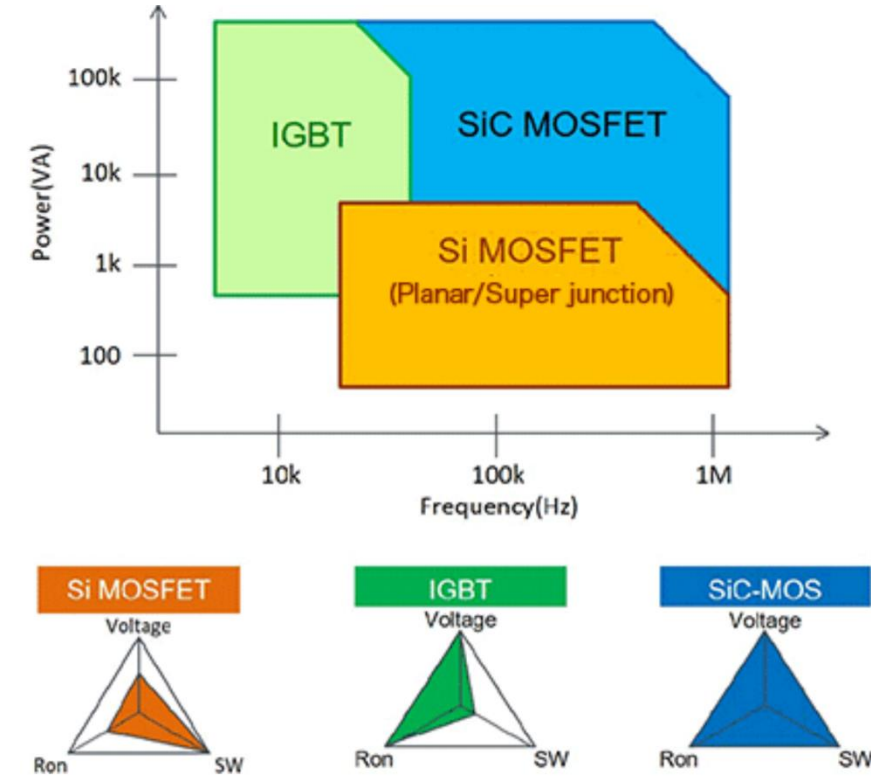
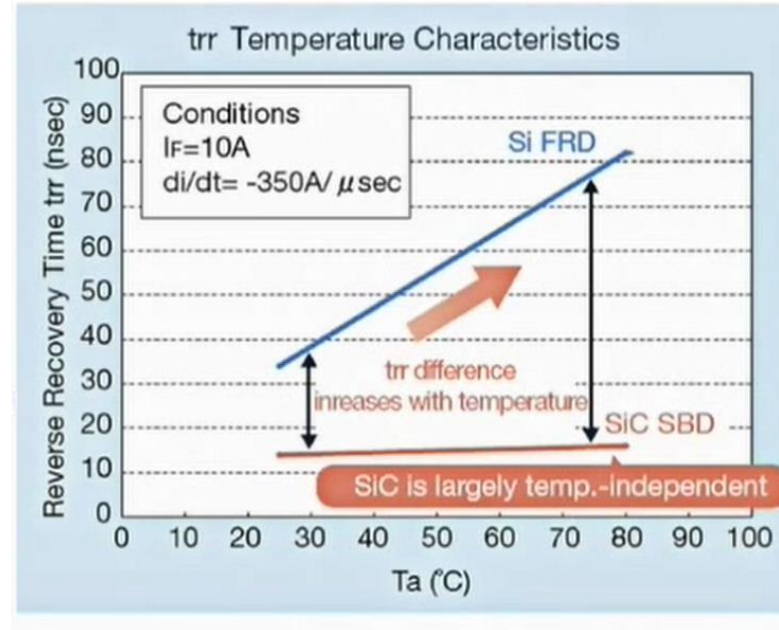
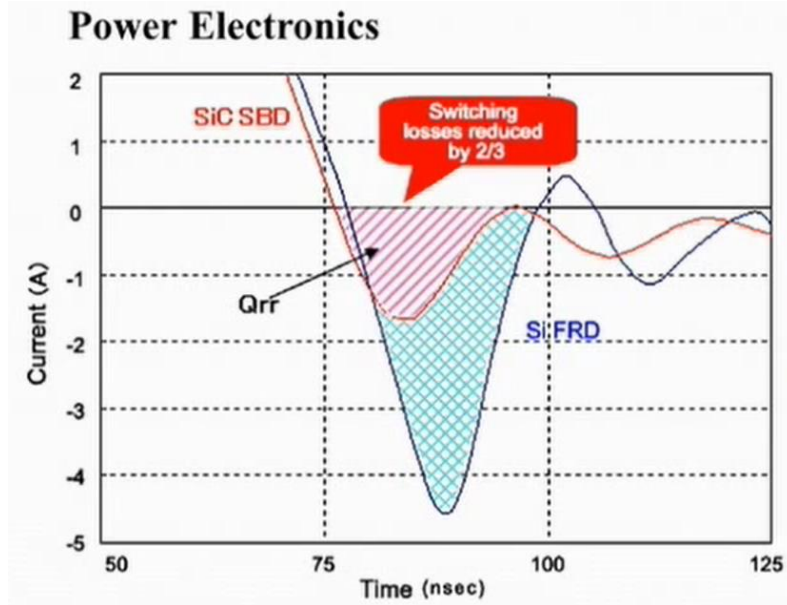


Contents in this Chapter

1. Properties of SiC
2. SiC power MOSFETs
3. Applications



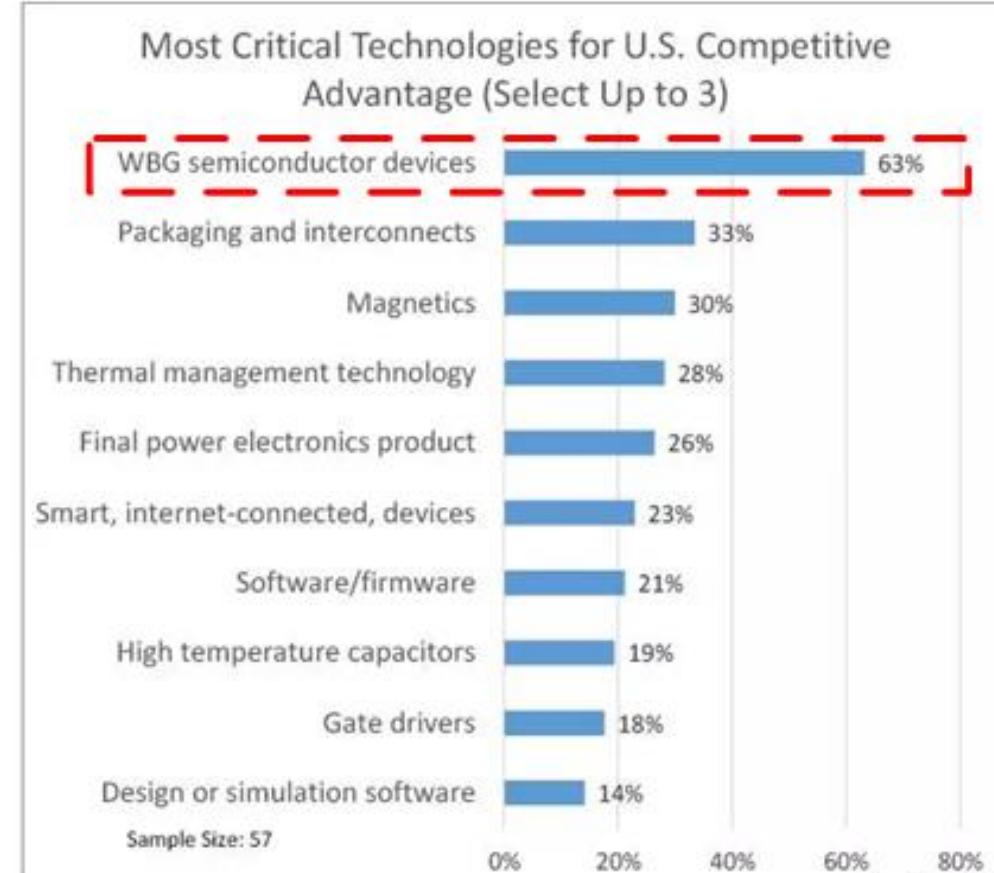
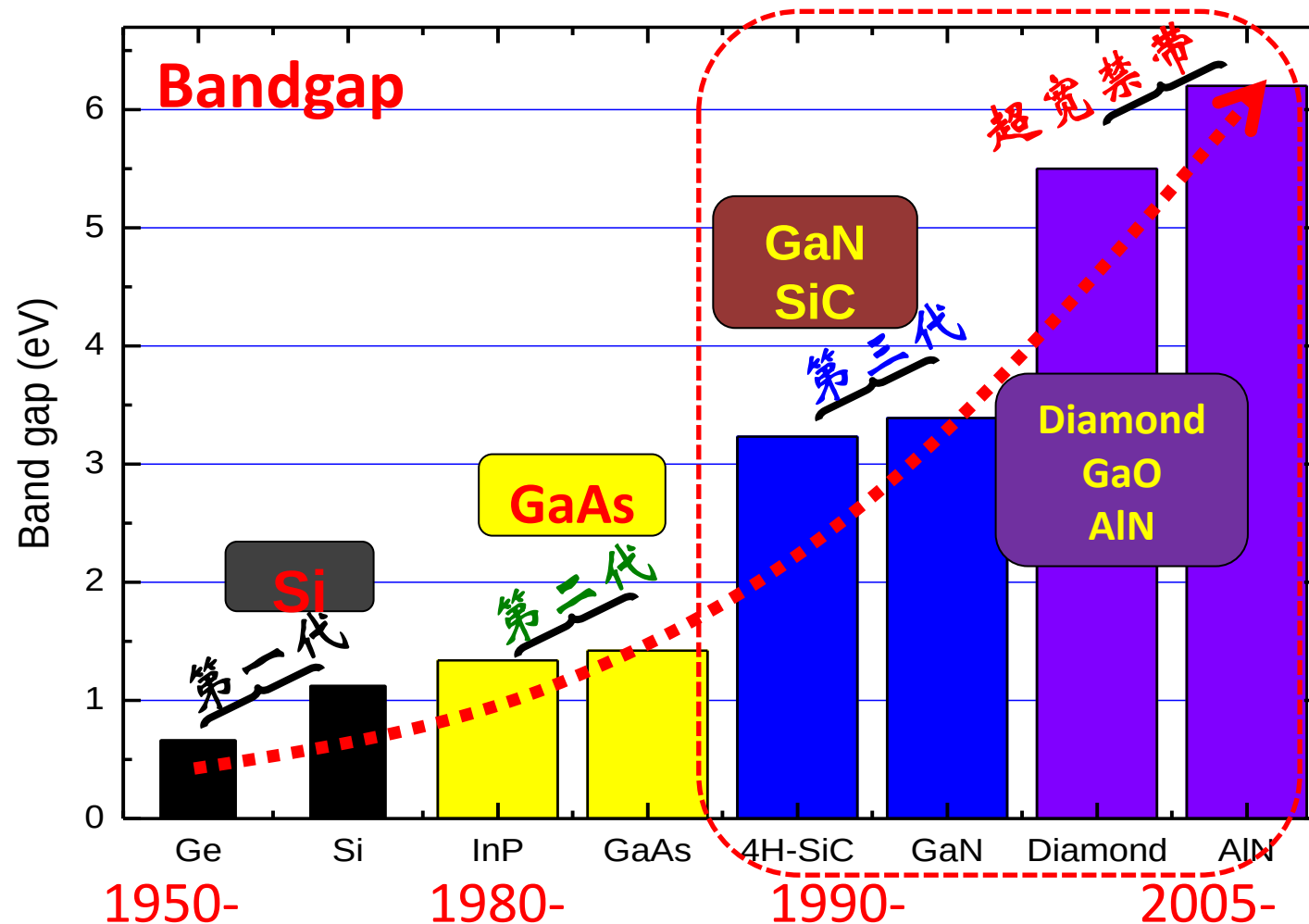
Why SiC ?



- Low loss
- High-efficiency power conversion
- Small heat sinks
- Reduced cooling costs

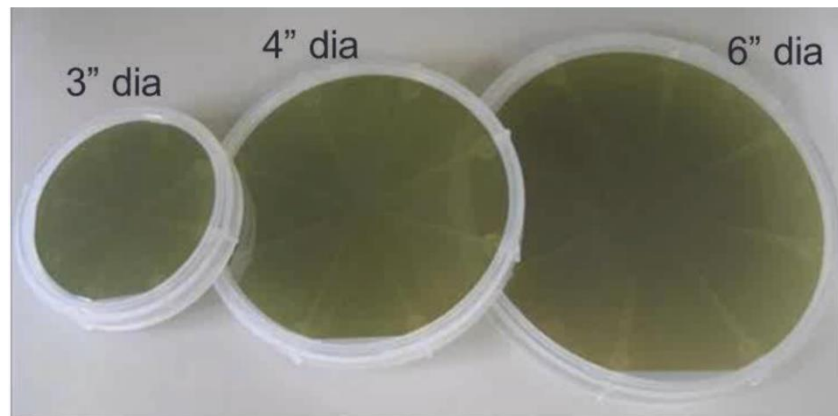
ROHM Semiconductor, White paper (2015)

Era of 3rd generations of semiconductors



SiC technology

- Most advanced technologies compared to other wide bandgap materials
- SiC not found naturally in earth, discovered in 1880s
- Polycrystalline powder used widely for abrasive applications
- Major advances in single crystal wafers for electronics starting in late 80s
- Wafer development with continual diameter increase and reduction of the cost



http://www.sdk.co.jp/english/news/2008/aanw_08_0992.html

Silicon wafer : 12 inch → 18 inch
SiC is a “teenager” compared to Si



Wide bandgap semiconductor R&D in US

National Network for Manufacturing Innovation

In January 2013, the Administration [released a report](#) that details the President's proposal for a National Network for Manufacturing Innovation.

Technology Areas for New Institutes:

- **Digital Manufacturing and Design Innovation:** Advanced design and manufacturing tools that are digitally integrated and networked with supply chains can lead to 'factories of the future' forming an agile U.S. industrial base with significant speed to market advantage.
- **Lightweight and Modern Metals Manufacturing:** Advanced lightweight metals possess mechanical and electrical properties comparable to traditional materials while enabling much lighter components and products.

• **Next Generation Power Electronics:** Wide bandgap semiconductor based power electronic devices represent the next major platform beyond the silicon based devices that have driven major technological advances in our economy over the last several decades. Wide bandgap technology will enable dramatically more compact and efficient power electronic devices for electric vehicles, renewable power interconnection, industrial-scale variable speed drive motors and a smarter more flexible grid; in addition to high-performance defense applications (e.g. reducing the size of a sub-station to a suit case).

The winning consortium, led by **North Carolina State University** and headquartered in Raleigh, North Carolina, includes the State of North Carolina and: **18 Companies:** ABB, APEI, Avogy, Cree, Delphi, Delta Products, DfR Solutions, Gridbridge, Hesse Mechantronics, II-VI, IQE, John Deere, Monolith Semiconductor, RF Micro Devices, Toshiba International, Transphorm, USCi, Vacon
7 Universities and Labs: North Carolina State [Lead], Arizona State University, Florida State University, University of California at Santa Barbara, Virginia Polytechnic Institute, National Renewable Energy Laboratory, U.S. Naval Research Laboratory

Source: www.whitehouse.gov

EE345 – Introduction to Wide-bandgap Semiconductors



Wide bandgap semiconductor R&D in China

- 第三代半导体材料及器件的突破将引发科技变革并重塑国际半导体产业格局。美、日、欧、韩等发达国家高度重视并已部署国家相关计划抢占战略制高点。我国政府也十分重视第三代半导体材料与器件的研发与产业化，在科技部等部委的持续部署支持下，我国材料研发的整体水平与国际上差距不大，**半导体照明作为在第三代半导体的材料首个第一个产业化的应用方向方面——半导体照明已经在关键技术上实现突破，创新应用处于国际领先水平**；此外，在第三代半导体电子器件应用方面，在移动通讯、光伏逆变、雷达等领域已有少量示范应用。
- 2015年9月9日，在国家科技部、工信部、北京市科委的支持下，由第三代半导体相关的科研机构、大专院校、龙头企业筹建“第三代半导体产业技术创新战略联盟”。
- 自2015年以来，国家科技部重点研发计划“战略性先进材料”专项中成立涉及第三代半导体方面的专项



我国从事宽禁带半导体行业的大学，研究院所，公司主要有：

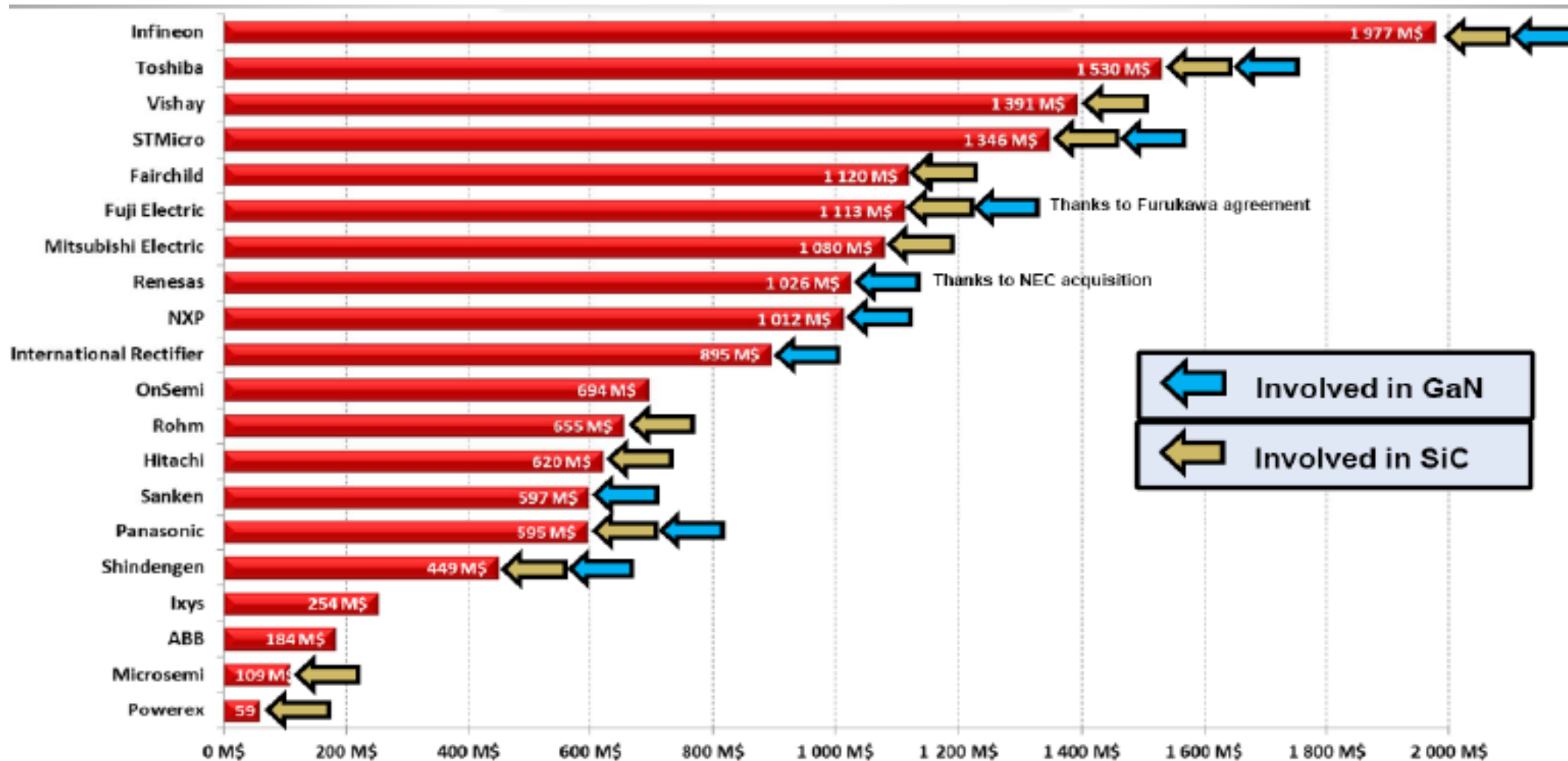
研究机构：北京大学，北京工业大学，南昌大学，山东大学，浙江大学，西安交通大学，西安电子科技大学，南京大学，中科院半导体所，中科院苏州纳米所，中国电科第十三所，南方科技大学等

公司：中兴通讯，中电科电子装备，广东德豪润达，苏州能讯，三安光电，晶能光电，株洲中车，东莞中稼半导体，苏州纳维，江苏华功半导体，杭州士兰微电子，易美芯光，晶湛半导体等

Source: www.casa-china.cn

www.sustc.edu.cn

Top 20 companies of Wide bandgap semiconductors



Source: Yole Développement



Market of SiC devices – 民用市场

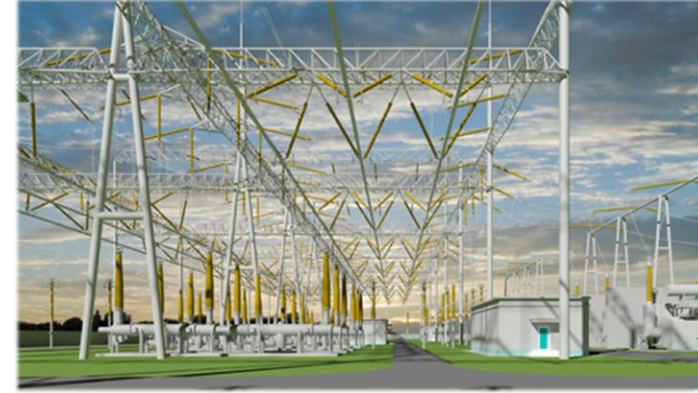


- SiC材料应用在高铁领域，可节能**20%**以上，并减小电力系统体积；
- 应用在新新能源汽车领域，可降低能耗**20%**；
- 应用在家电领域，可节能**50%**；
- 应用在大数据领域，可帮助数据中心能耗大幅降低(当前全球**300**万台数据中心每小时耗电量约为**3000**万千瓦)；





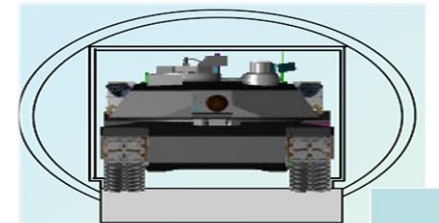
Market of SiC devices – 民用市场



- 应用在风力发电领域，可提高效率**20%**；
- 应用在太阳能领域，可降低光电转换损失**25%以上**；
- 应用在工业电机领域，可节能**30%-50%**；
- 应用在超高压直流输送电和智能电网领域，可使电力损失降低**60%**，同时供电效率提高**40%以上**；



Market of SiC devices – 军用市场



航母，驱逐舰以及潜艇的先进的固态变电站系统、电力转换和传输系统

混合动力战车电力驱动系统、战车电源系统、战机各种辅助电力驱动系统

激光武器、电磁弹射、电磁炮等新型武器系统所需的高能量高功率脉冲电源

简化电力电子变换模块和装置、提高性能，效率

卫星、太空探测器和空间站中的太阳能电池、动力驱动系统和能源控制转换系统

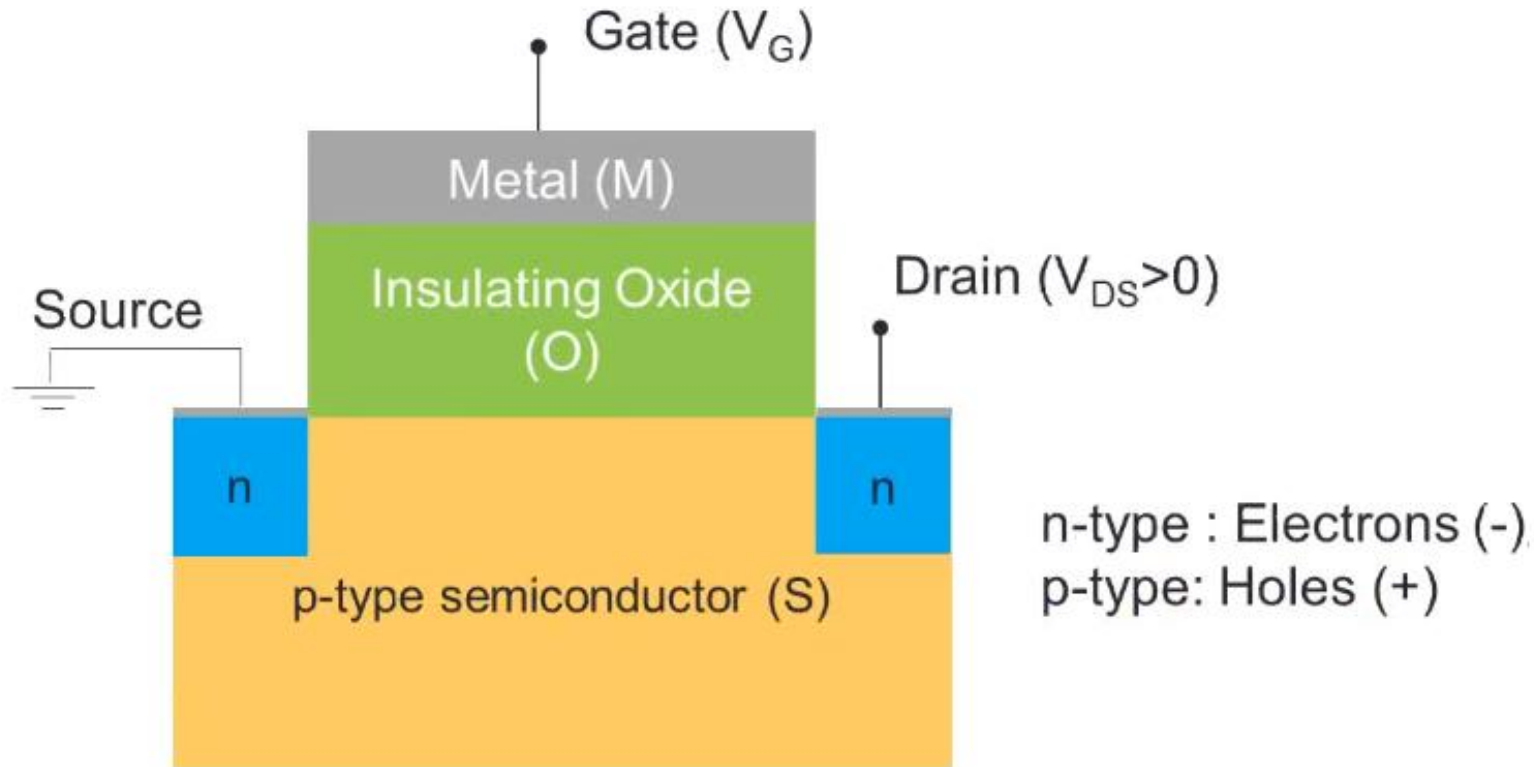
SiC高电压、低能耗、更快速度、高温工作能力、抗辐照

Example

美国Cree公司受军方资助，已开发出10kV/50A的SiC PiN整流器件和10kV的SiC MOSFET。下一步他们将要按比例缩小这些器件的尺寸，以得到10kV/110A的器件模块，并将它们用于航母的电气升级管理中，届时将节约航母重量170吨，节约体积290立方米。美国更重视SiC器件在军事方面的应用。



How does a MOSFET work?



Key parameters of a MOSFET:

W/L

T_{ox}

C_{ox}

Mobility

V_{th}

I_{on}

I_{off}

ON/Off ration

Subthreshold slope (SS)

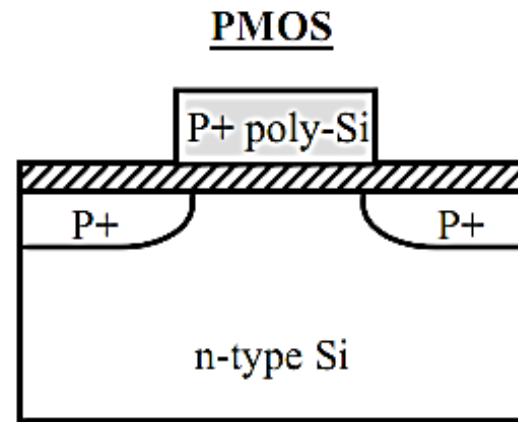
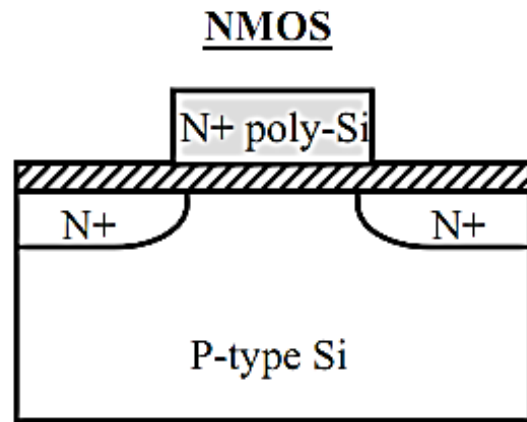
R_{on}

Anything more for SiC MOSFET?

Breakdown voltage

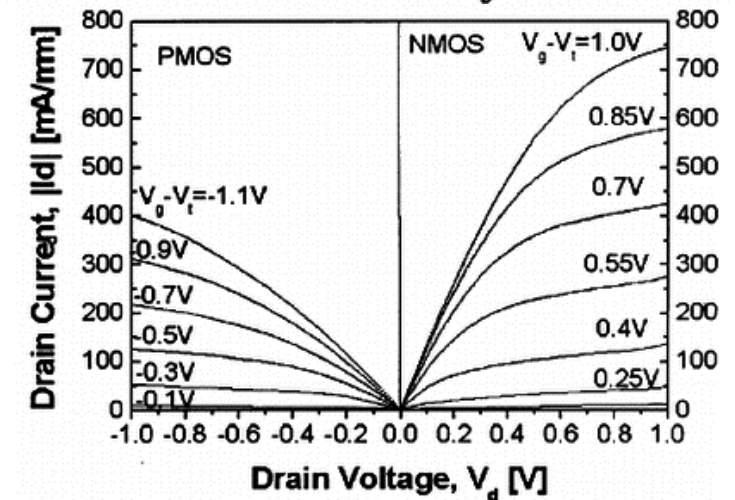
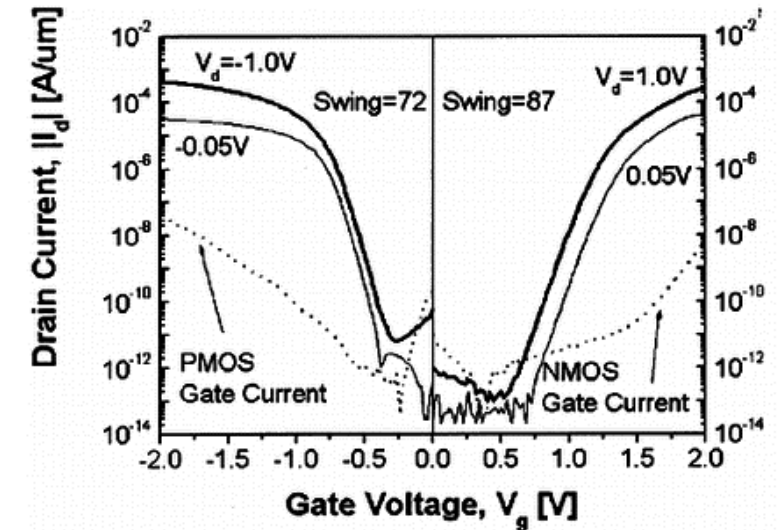
Thermal conduction coefficient

N-type and P-type MOSFET



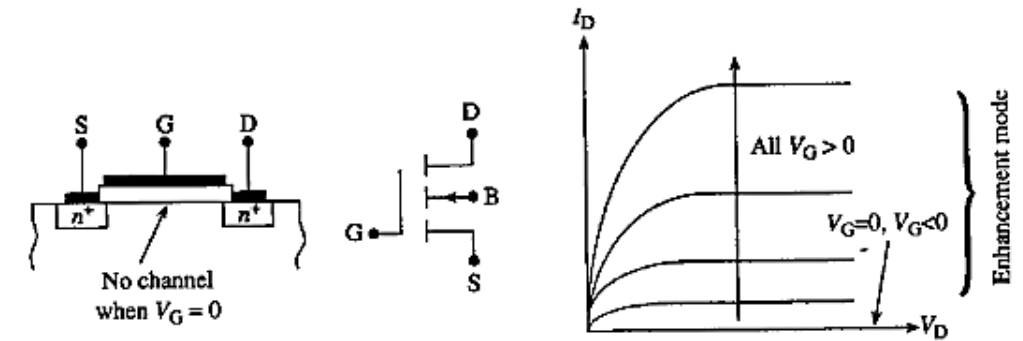
- For current to flow, $V_{GS} > V_T$
- Enhancement mode: $V_T > 0$
- Depletion mode: $V_T < 0$
 - Transistor is ON when $V_G = 0V$

- For current to flow, $V_{GS} < V_T$
- Enhancement mode: $V_T < 0$
- Depletion mode: $V_T > 0$
 - Transistor is ON when $V_G = 0V$

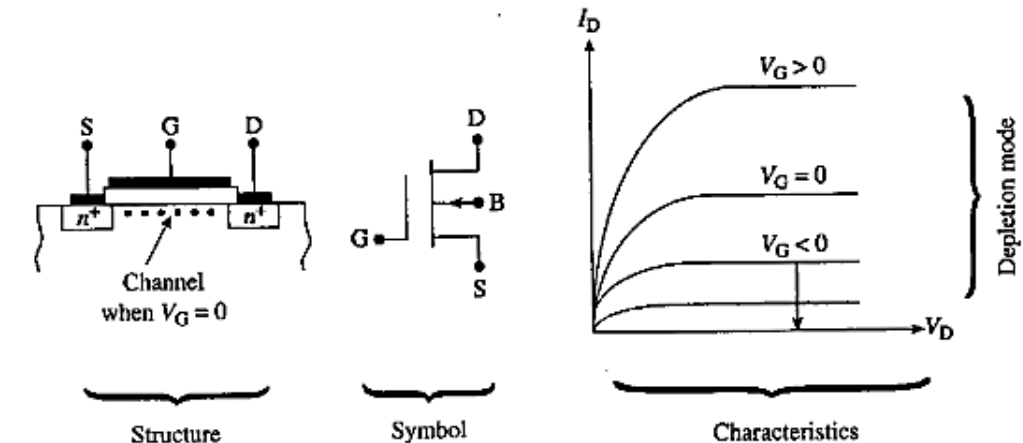


Enhancement mode and depletion mode

- **Enhancement mode: normally-off**, gate bias is needed to connect the source and the drain (turn on the transistor).
- **Depletion mode: normally-on**, a gate bias is needed to block the conduction between the source and the drain (turn off the transistor).



Enhancement Mode MOSFET

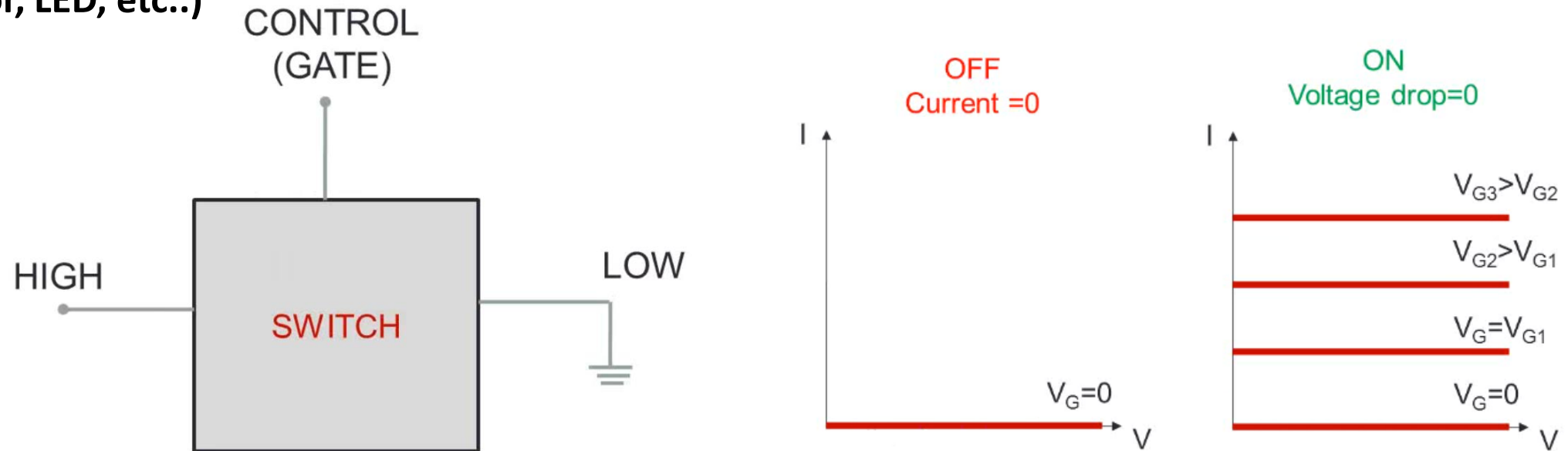


Depletion Mode MOSFET

Homework question: 对于集成电路来说, 耗尽型器件和增强型器件哪一种更好, 为什么, 举例说明

How does a power switch work?

Power switch is a device used to control power flow from power supply to load (Resistor, Capacitor, inductor, LED, etc..)



AN IDEAL POWER SWITCH:

- ✓ Block High Voltage (Infinite resistance in OFF State)
- ✓ Conduct High Current (Zero resistance and voltage drop in ON State)
- ✓ Without ANY Power Dissipation



Classification of SiC Power devices

- **SiC Schottky Diode:** the first successfully commercialized SiC power device. Majority carrier device. Breakdown voltage above 1000V (Si is below 200V). Forward voltage drop is 1-2V (Si is 0.7V)
- **SiC JFET:** was also commercialized earlier than other SiC power devices. No reliability issue of gate oxide (Comparing with SiC MOSFET). Basically a normally-on device. Low controllability. Typically connected with a Si MOSFET for a cascade configuration.
- **SiC BJT:** theoretically can provide the lowest specific ON-resistance that results in the smallest chip size and parasitic capacitance for a given voltage and current rating. Allowing higher switching speed than other SiC devices. It has no oxide layer so can operate at higher temperature. It is a current controlled device. Needs a certain amount current to maintain ON status.
- **SiC MOSFET: voltage controlled and normally OFF device.** has traditionally suffered from poor oxide-semiconductor interface quality which led to large threshold voltage instability. The ON-resistance increase 20% when the junction temperature increased from 25 C to 135 C (Si MOSFET increase 250%). The zero gate leakage current does not increase much with temperature (Si MOSFET gate leakage current increase 100 times when junction temperature from 25 C to 135 C)





Evolution of power electronics -1

➤ Si功率半导体器件的发展历程及限制

历程---经历了如下三代:

第一代-----Si双极晶体管(BJT)、晶闸管 (SCR) 及其派生器件。

优点： 工作电流大，适合低频大功率应用。

缺点： 工作频率仅在在KHz范围；（工作频率低，自身损耗大）

需要大的基极驱动电流; 电流增益 β 小；器件的并联使用困难；

$I-V$ 的负阻特性引起的二次击穿，使安全工作区（SOA）受到限制。





Evolution of power electronics -2

第二代-功率MOSFET

优点：输入阻抗极高，（栅）控制电流极小；多子器件，可工作在100KHz以上的频率下；安全工作区大。从上世纪80年代初期开始得到迅速的发展，已形成了大量产品，并且在实际中得到广泛的应用。

缺点：导通电阻 r_{ON} （损耗）以及跨导 g_m 比双极器件以更快的速率随击穿电压增加而变坏，使它们在高压工作时损耗大。





Evolution of power electronics -3

第三代-绝缘栅双极晶体管 (IGBT)

MOSFET同双极晶体管的复合器件，兼有二种器件的优点，高压时导通电阻小，工作频率比双极的高，适合于中、高压大功率应用。自1982年由美国GE公司提出以来，发展十分迅速。

商用的高压大电流IGBT器件仍在发展中，尽管德国的EUPEC最早生产的6500V/600A高压大功率IGBT已经获得实际的应用，但其电压和电流容量还不能完全满足电力电子应用发展的需要，特别是在高压（10KV）以上的应用领域中，目前还只能通过IGBT串联技术来实现。





Main limitations of Si power devices

Si功率器件受限原因:

限制主要由Si材料的性质引起:

- * Si的临界击穿场强 E_c 小, 限制了器件的最高工作电压以及导通电阻, 它使Si功率半导体器件的开关损耗难以达到理想状态。

- * Si较小的带隙 E_g 及较低的热导率 λ , 限制了器件的最高工作温度($\sim 200^\circ\text{C}$)以及最大功率。

为了满足不断发展的电力电子工业的需要, 以及更好地适应节能节电的大政方针, 需要发展新半导体材料的功率器件。SiC以及GaN功率器件正是适应这一形势需要发展起来的。



Merit of SiC devices

➤ SiC功率半导体器件的优点

优质指数

一个功率器件的比导通电阻为: $R_{\text{on-sp}} = (W_D / q\mu_n N_B)$

W_D : 器件有源区耗尽区长度, $W_D = 2V_B / Ec$

N_B : 有源区掺杂浓度: , $N_B = \epsilon_s Ec^2 / 2qV_B$

∴ 比导通电阻:
$$R_{\text{on-sp}} = 4V_B^2 / \epsilon_s \mu_n Ec^3 \quad (1)$$

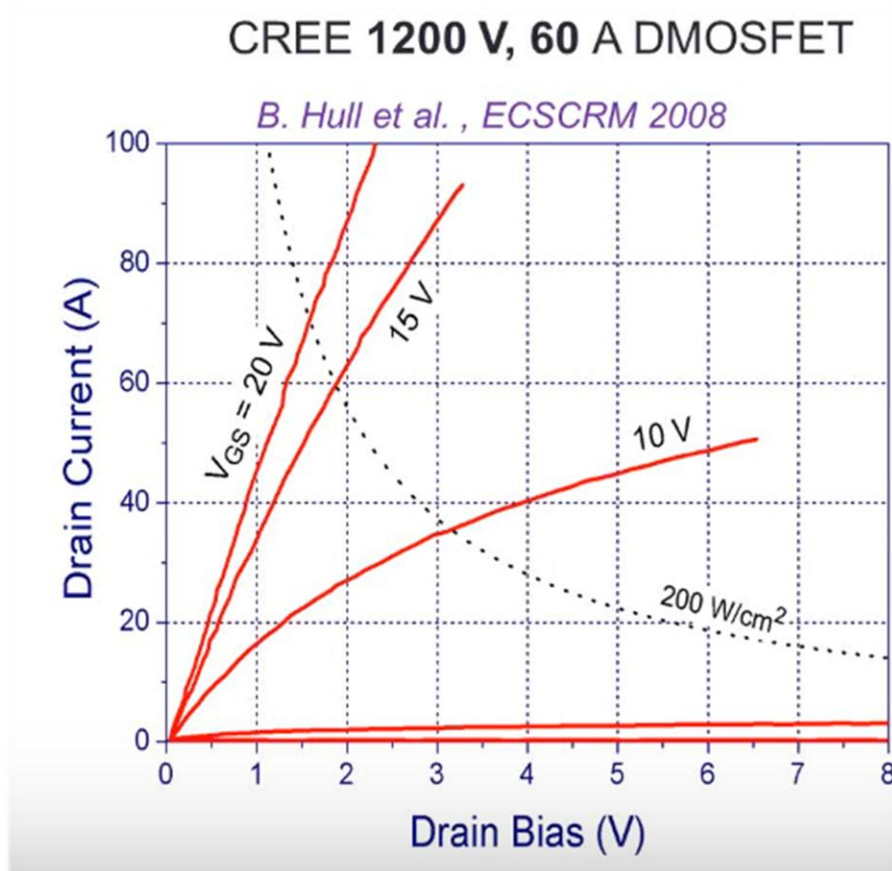
定义优质指数BFOM (Baliga's figure of merit)

$$= \epsilon_s \mu_n Ec^3 \quad (\text{导通电阻越小, 优质系数大}) \quad (2)$$

W_D : 器件有源区耗尽区长度; V_B : 击穿电压; μ_n : 载流子迁移率; N_B : 掺杂浓度; Ec : 材料临界电场强度; ϵ_s : 材料介电常数



Example: SiC Power MOSFET



- ✓ $R_{on} = 6-8 \text{ m}\Omega \cdot \text{cm}^2$
- ✓ Ideal $R_{on} = 1-2 \text{ m}\Omega \cdot \text{cm}^2$
- ✓ Significantly better than Si but channel resistance/mobility limits performance
- ✓ Lower R desirable for smaller, faster and more energy efficient power switches

Question: How to calculate R_{on} ?



Conductivity – mobility relationships

$$\text{Conductivity } \sigma = n \cdot q \cdot \mu.$$

where n = number of electrons per unit volume
 μ = mobility and q = elemental charge of e^-

$$\text{Resistivity } \rho = 1 / \sigma$$

Units: [Ohm.cm]

$$\text{Resistance } R = \rho l / A$$

Units: [Ohm]

where l = length of material A = Cross-sectional area



Electron transport in semiconductors: Mobility

- Mobility (μ) dictates how fast electrons can move in a material under an applied electric field

$v = \mu \cdot E$ where v = velocity, E = Electric field

Units: [$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$]

- Defined alternatively:

$\mu = q\tau/m^*$ where q = charge of $e^- = 1.6 \times 10^{-19} \text{ C}$

τ = Average time between scattering events i.e. how long the e^- is accelerated by E before it collides with something that changes direction or energy

m^* = Effective mass of e^- in a solid

- Matthiessen's rule: When more than one scattering processes are present $1/\tau = 1/\tau_1 + 1/\tau_2 + 1/\tau_3 + \dots$

$$1/\mu = 1/\mu_1 + 1/\mu_2 + 1/\mu_3 + \dots$$

where μ_i = mobility associated with the i^{th} scattering process

Process with lowest mobility limits the total mobility

(Just like resistors in a parallel connection: Lowest value dominates !!)

Classification of SiC devices

SiC devices:

Schottky diodes

PIN/JBS

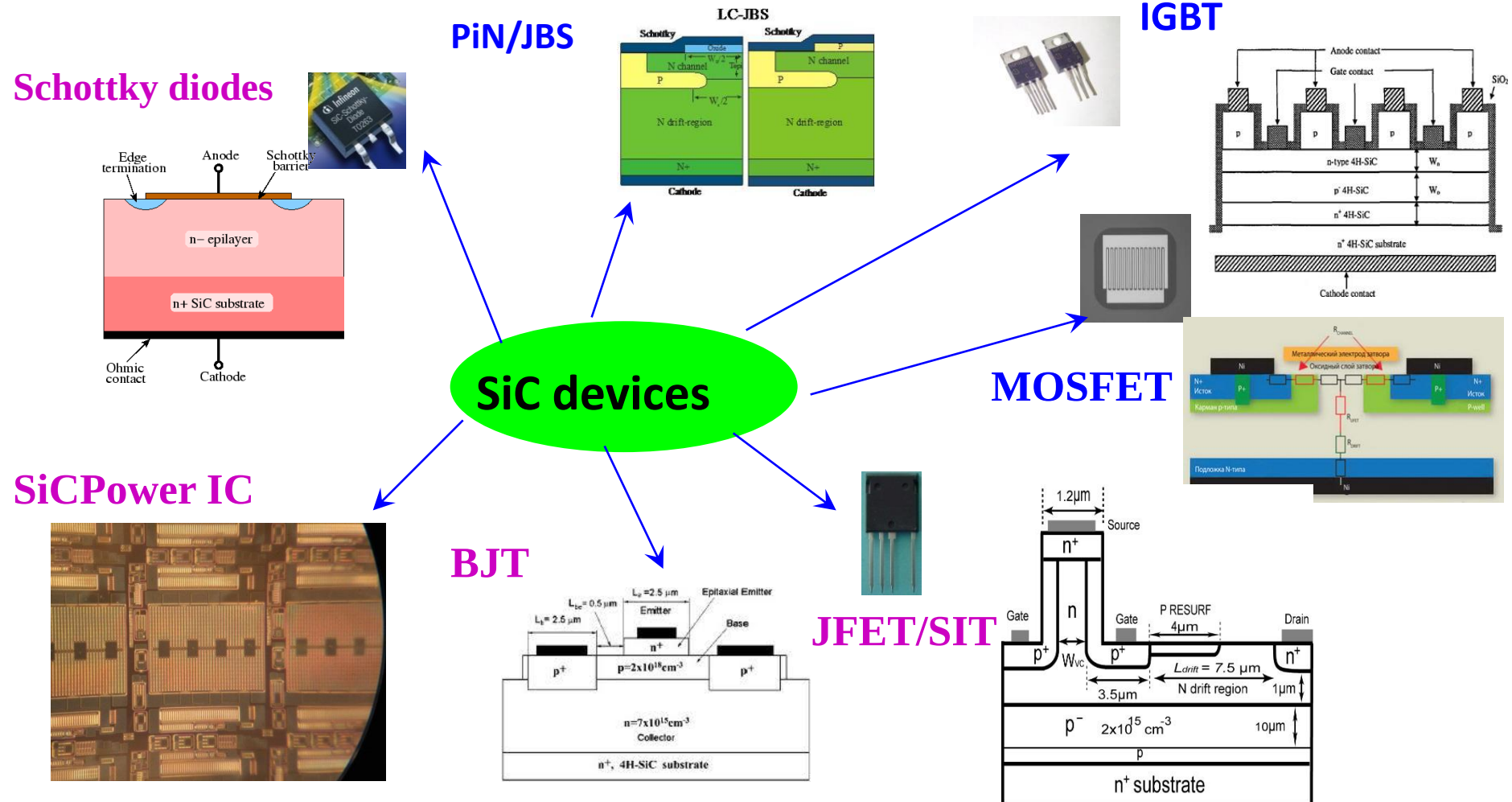
IGBT

MOSFET

BJT

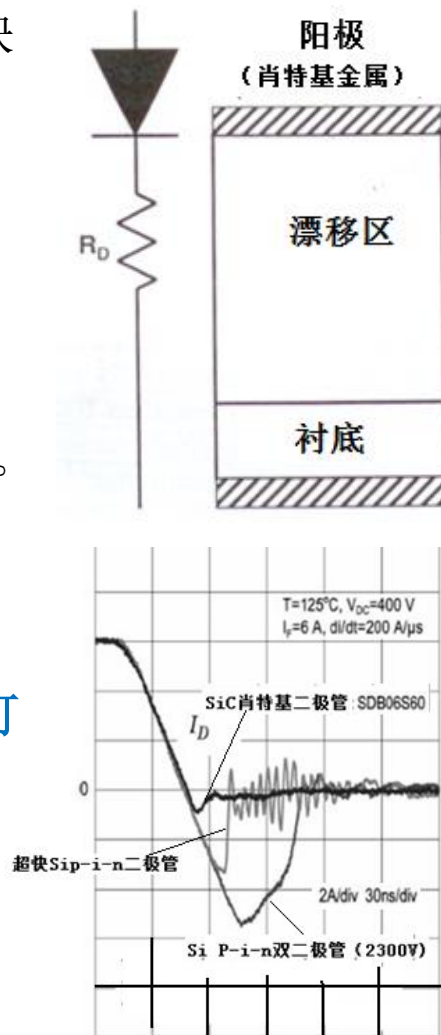
JFET

Power IC

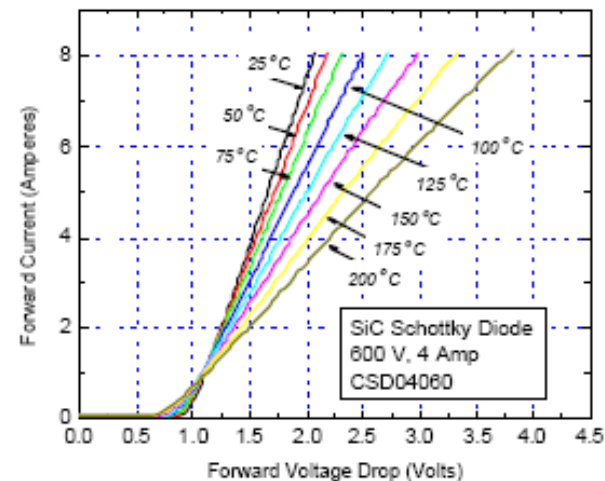


Schottky diodes

肖特基二极管是单极器件，具有快的正向到反向的恢复时间（右下图），是电力电子中重要的高频整流元件。对于Si器件，在较高击穿电压时漂移区电阻迅速增加，由此功率损耗显著。**Si肖特基二极管工作电压一般约为200V**，改进的结构也难以超过600V。**SiC肖特基二极管可以用低得多的漂移区厚度获得很高的击穿电压。**



- Cree offers a family of silicon carbide (SiC) based ZERO RECOVERY® Rectifiers.
- 300V, 600V and 1200V breakdown
- Current ratings of 10A and 20A at 300V;
- 1A, 4A, 6A, 10A and 20A at 600V
- and 5A, 10A and 20A at 1200V.
- Extremely fast switching (<50 ns)
- Temperature independent switching behavior
- Positive temperature coefficient on forward voltage
- High frequency operation.

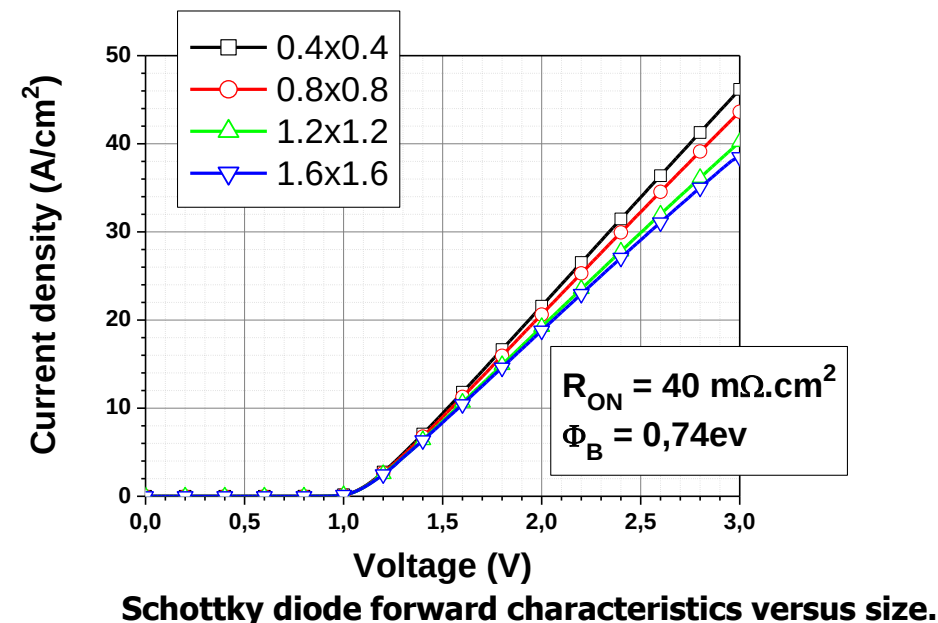
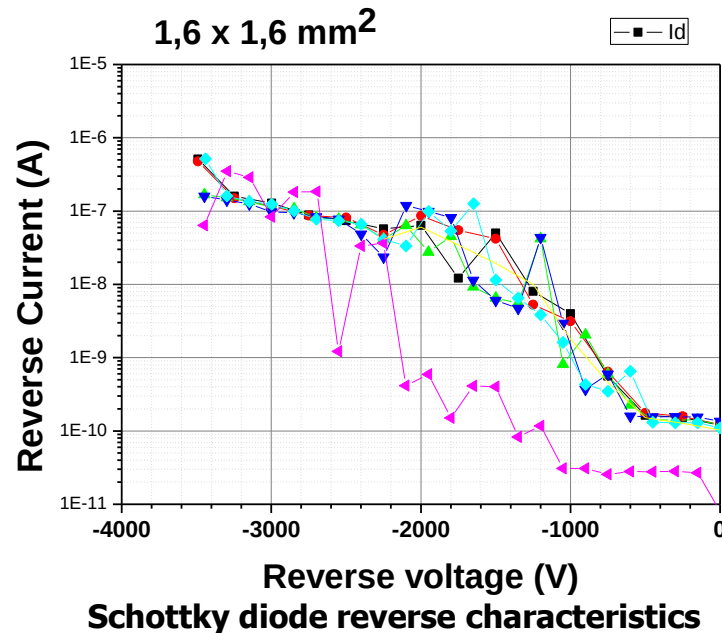


Forward characteristics of a 4A/600 V SiC SBD.

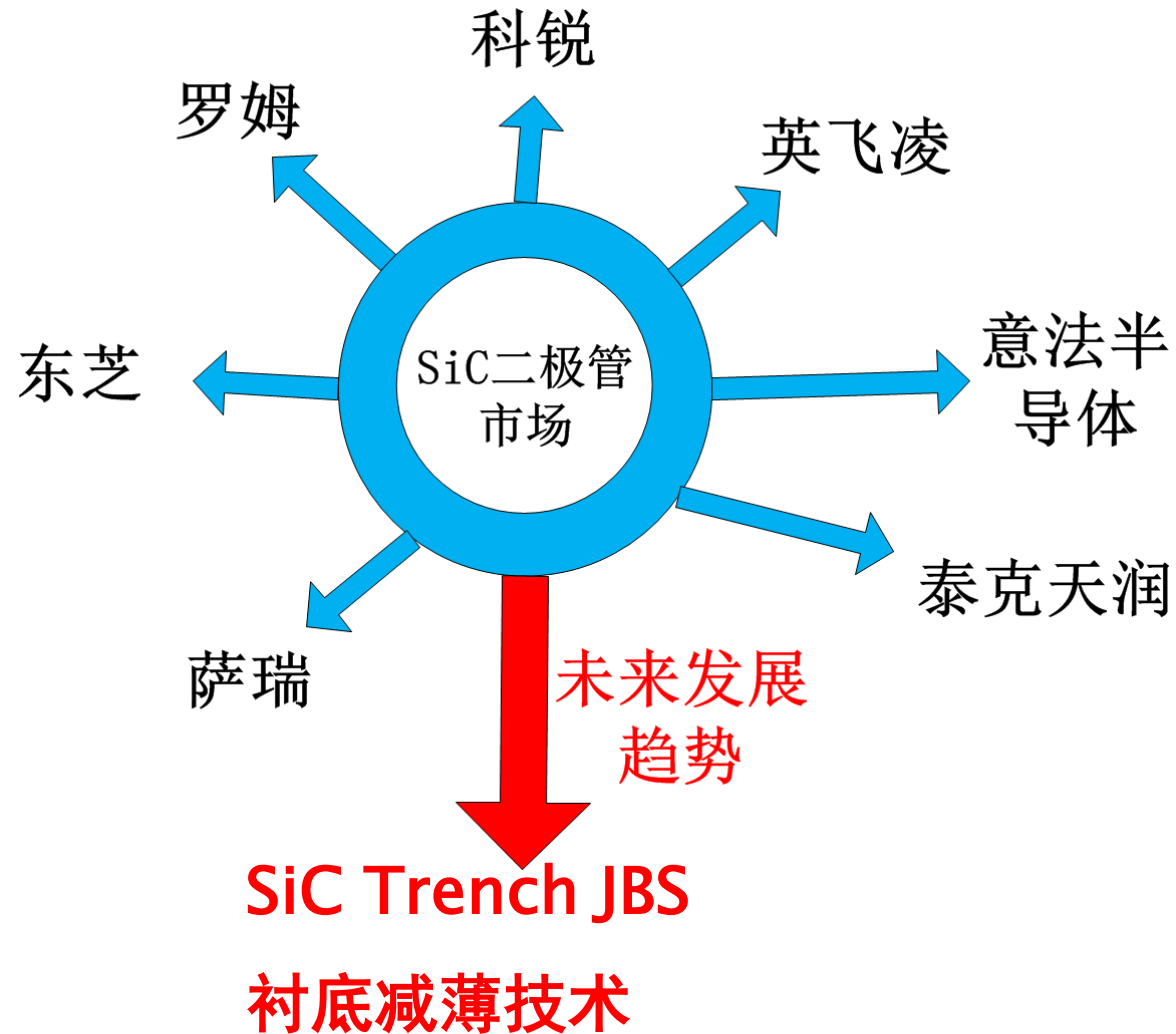
I-V characteristics of Schottky diodes

➤ 3.5kV Diodes fabrication for module

- Good current density uniformity vs diodes size
- $R_{ON} = 40 \text{ m}\Omega \cdot \text{cm}^2$ close to theoretical expected value ($31 \text{ m}\Omega \cdot \text{cm}^2$).
- Very low reverse leakage current density ($J_R < 10 \text{ }\mu\text{A}/\text{cm}^2$ @ 3.5kV)
- 4.7kV Breakdown voltage measured = termination efficiency of at least 80%



Current status of SiC Power Diodes



二极管种类:

600V(1~24A)

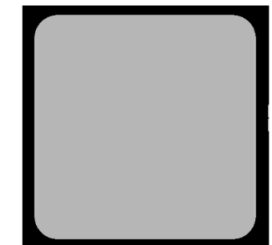
650V(2~50A)

1200V(2~50A)

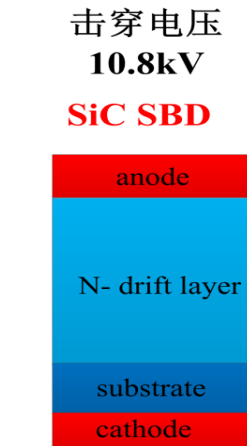
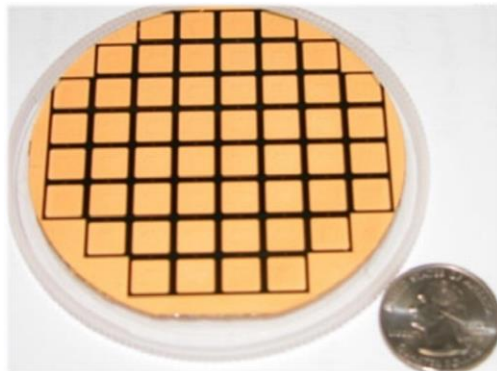
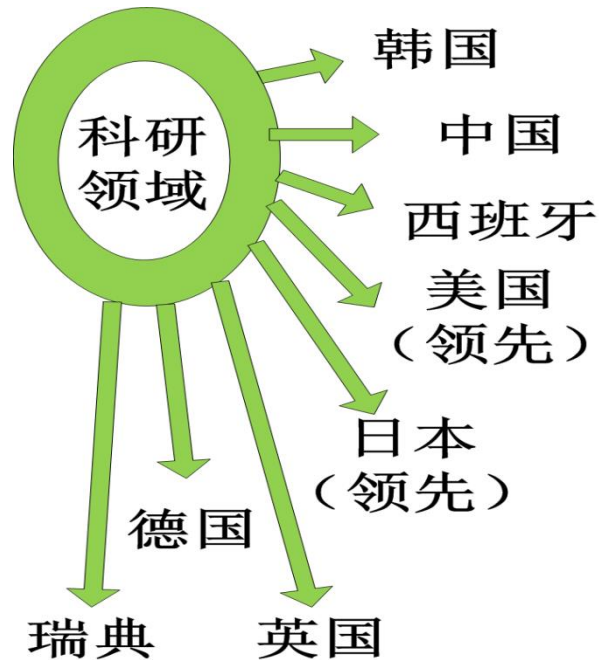
1700V(4~30A)

主流结构:

SiC JBS二极管

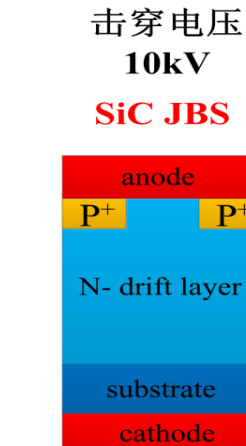


Trend of SiC power diodes



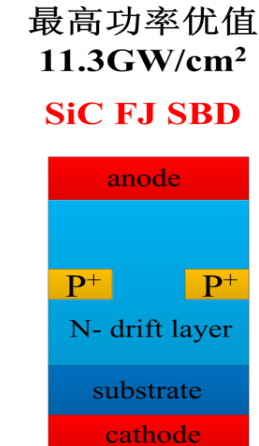
★克服表面缺陷，实现低压大电流器件

e. g.
2014年韩国：
20A/600V



★具有沟槽有源区的SiC JBS器件

e. g.
Cree5代产品



★改善开关特性

e. g.
2009年日本：
SiC FJ-PiN



★提高材料质量，实现超高压器件

e. g.
2014年德国：
1000A/6500V

SiC FET

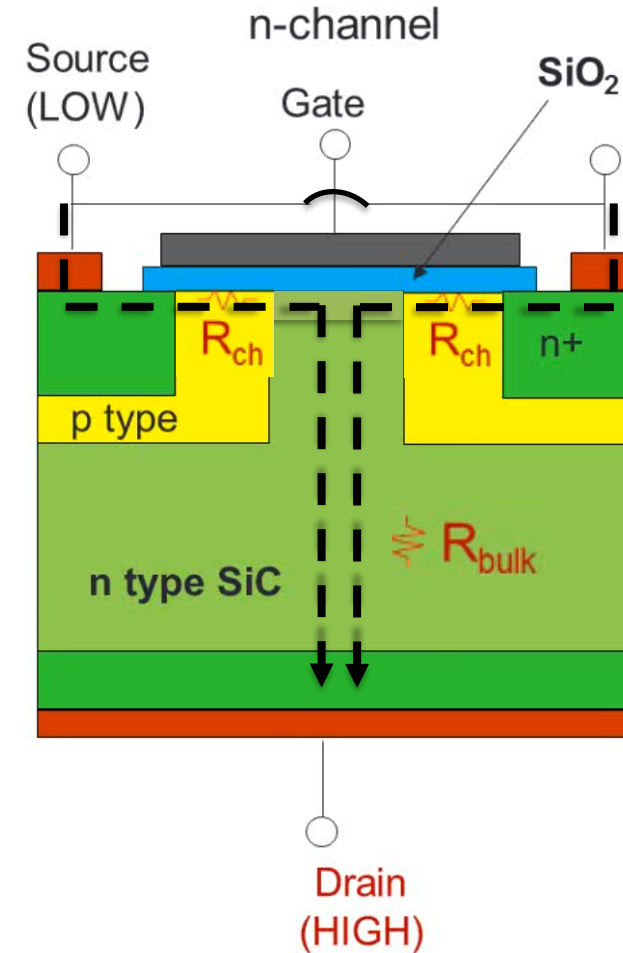


Homework question: What is FET? JFET? MESFET? MISFET? MOSFET?

Device structure of a typical SiC power MOSFET

- Three-terminal n-channel device
- Vertical-structure device: Gate and source terminals on the top. Drain terminal at the bottom.
- With a thick layer of lightly-doped n-type SiC to block high voltage
- The thickness and doping concentration of the blocking layer are designed to meet the requirements of the blocking voltage
- Gate voltage
- When the transistor is ON, two effective resistances: Channel resistance and bulk resistance

$$R = R_{\text{channel}} + R_{\text{bulk}}$$



Relationship between resistance and mobility

R_{bulk} inversely proportional to the electron mobility in the **bulk semiconductor**

$$R_{bulk} \propto \frac{1}{\mu_{bulk}}$$

$R_{channel}$ is inversely proportional to the electron mobility in the **MOSFET channel**

$$R_{channel} \propto \frac{1}{\mu_{channel}}$$

Comparison between bulk resistance of Si and SiC

$$R_{\text{bulk}} = \frac{d}{q\mu_{\text{bulk}}N_d} \quad N_d = \frac{2\epsilon_s(E_C d - V_{BD})}{qd^2}$$

Material System	E_C (MV/cm)	μ_{bulk} (cm ² /V-s)	d (μm)	N_d (cm ⁻³)	R_D (mΩ-cm ²)
Silicon	0.2	1500	90	9.7E13	390
SiC	2	600	9	8.2E15	1.1

SiC Major Advantage:

$$\frac{R_{\text{bulk SiC}}}{R_{\text{bulk Si}}} \approx \frac{1}{355} \bigg|_{V_{BD}=1.2kV}$$

Where

R_D = specific resistance (Ω-cm²)

BV_{BD} = breakdown voltage (V)

d = thickness

N_d = number of donors per cm³

E_C = critical electric field (V/cm)

μ_{bulk} = electron mobility in bulk

ϵ = material permittivity

Comparison between channel resistance

$$R_{channel} \propto \frac{1}{\mu_{channel}}$$

Channel Mobility Comparison

- $\mu_{channel\ Si} \approx 650\text{ cm}^2/\text{V-s}$
- $\mu_{channel\ SiC} \approx 20\text{-}35\text{ cm}^2/\text{V-s}$
- Critically depends on the properties of the oxide-semiconductor interface !

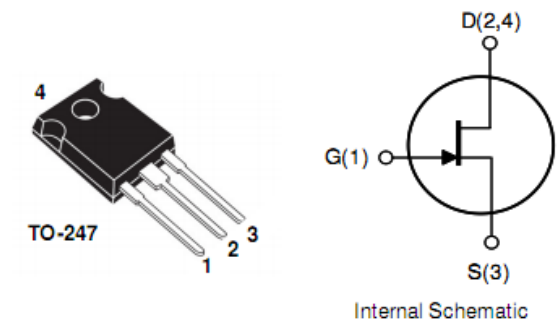
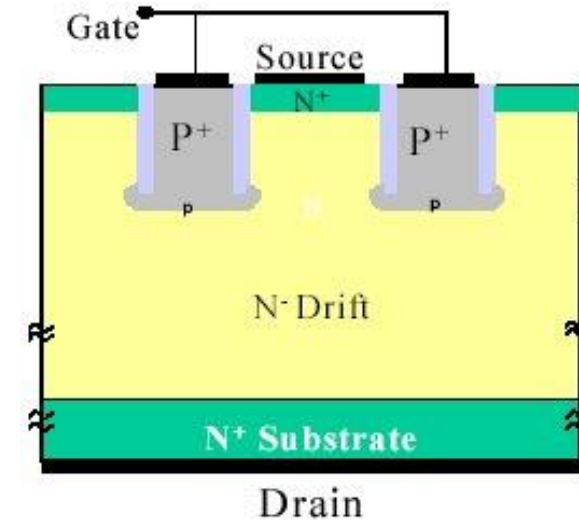
$$\frac{R_{chSiC}}{R_{chSi}} = \frac{\mu_{chSi}}{\mu_{chSiC}} \approx \frac{32}{1}$$

Resistance Summary:

- In Si, $R_{channel}$ is small compared to R_{bulk}
- In SiC, $R_{channel}$ is comparable to R_{bulk}

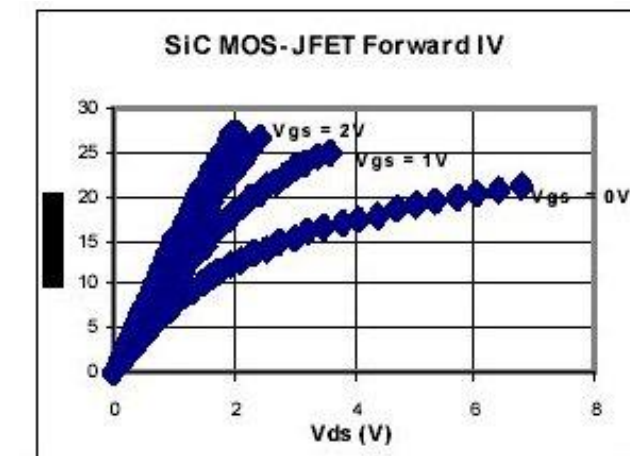
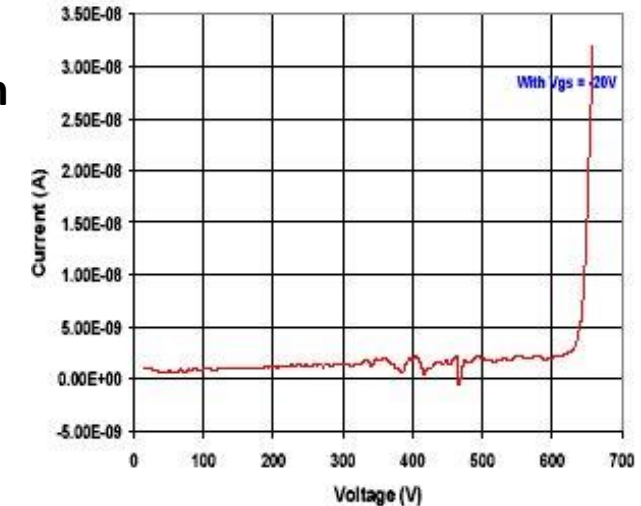
Recent progress of SiC FET -1

- Recent FETs have BVs greater than 4.5kV and R_{ons} of around 1/230 the theoretical limit for Si.
- However, their maximum controllable current is limited by the small sizes of chips due to the presence of crystal defects such as micropipes.
- In order to achieve higher current capacity new bipolar devices such as the 5kV SIJFET and 12.5kV SICGT modules have been developed.



Recent progress of SiC FET -2

- Newer line of commercial SiC MESFETs.
- Cree - selling 60 W MESFETs for RF applications that require a large bandwidth, such as military radios and jammers, and 10 W MESFETs for WiMAX applications.
- Reliability of devices - when failure is defined as a 20% change in any parameter, MESFET reliability is 250 years for an operating temperature of 175 ° C, and more than 7000 years for 225 ° C. *"GaN can't even come close to this in terms of reliability"* - Palmour.
- The 60 W MESFETs can be employed in broadband amplifiers to deliver 12 dB from 0.85-1.6 GHz at an average power-added-efficiency of 42%, when biased at 48 V. The device linearity is 10 dB higher than that of silicon LDMOS, which ultimately leads to efficiency improvements for wireless base station transmitters.



Recent progress of SiC FETs – 3

时间	机构	V_{DS}	I_D	类型
2005	Rutgers	14kV	33.9A/cm ²	常关
		1.7kV	98A/cm ²	
2008	U.S. ARL	1680V	54A	常关
		2055V	25A	
2013	南京电子器件研究所	1700V	3.5A	常开与常关
2014	浙江大学	3500V	15A	常关

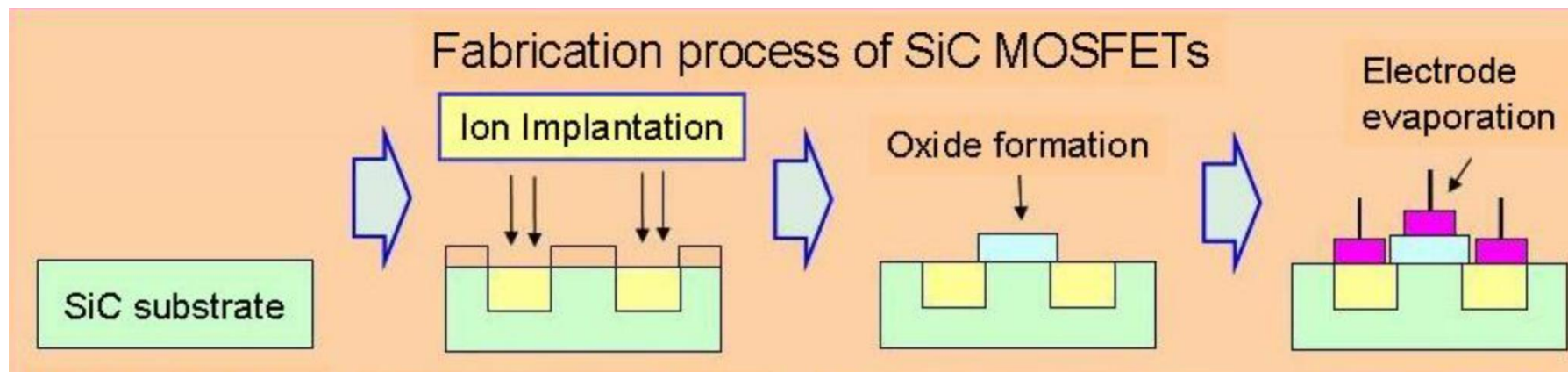
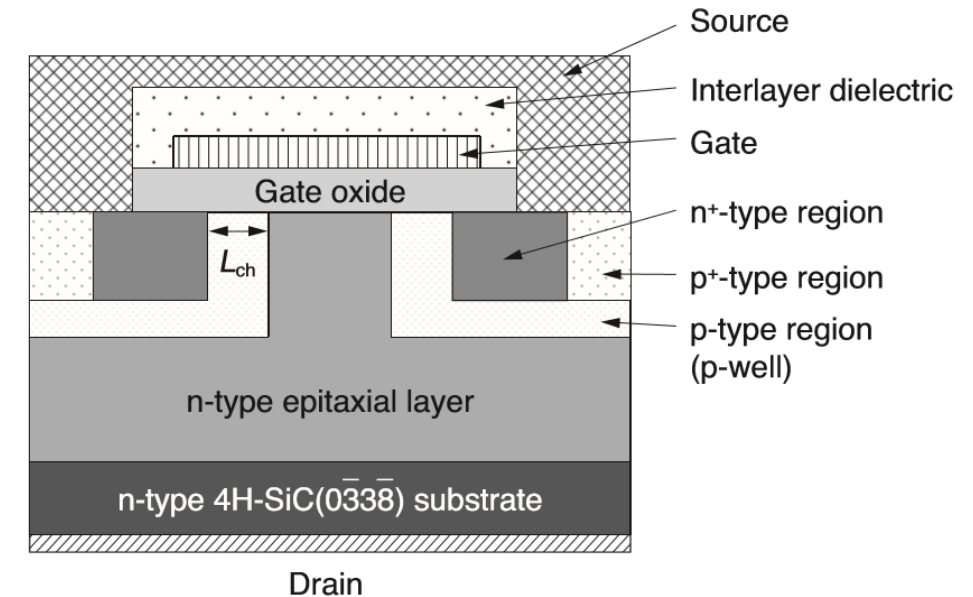
英飞凌与USCI公司已有几款代表性的产品面世，其性能参数如下：

机构	V_{DS}	I_D	类型
英飞凌	1200V	26~35A	常开
USCI	1200V	20~35A	常开



Fabrication process of typical SiC MOSFET

1. Epitaxial layer growth on substrates
2. P-type and n-type region formation by Al and P ion implantation (doping concentration of p-well can control threshold voltage)
3. Gate oxide formation by dry oxidation followed by nitridation and Ar annealing as post oxidation annealing (POA)
4. Gate electrode (usually Poly-Si or Al) formation by LPCVD
5. Metallization for Gate, Source, Drain electrodes





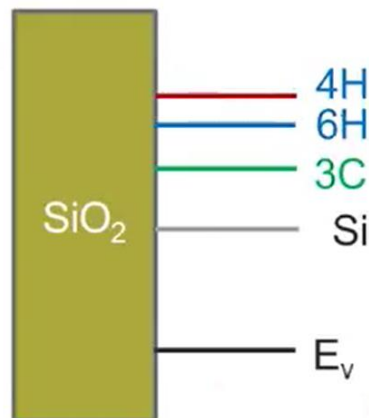
Critical Fabrication Techniques

- Photolithography (contact, contactless...)
- Thin film deposition (PECVD, LPCVD, sputtering, e-beam evaporation, thermal deposition, pulse laser deposition...)
- Annealing (furnace, RTA: rapid thermal annealing)
- Oxidation (dry oxidation, wet oxidation, natural oxide formation)
- Doping (diffusion, ion implantation)
- Metallization



SiO₂-SiC: A unique platform for interface engineering of power devices

- SiO₂-Si: the most widely studied material combination
- SiO₂-SiC: Only other semiconductor to thermally grow SiO₂
- Multiple polytypes (4H, 6H, 3C, ...), same chemistry but different crystal structures and band gaps
- Si-face (Conventional), C-face and non-polar faces
- Provides new controls to study MOS interface

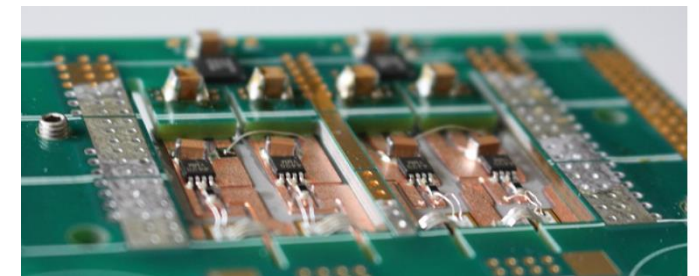
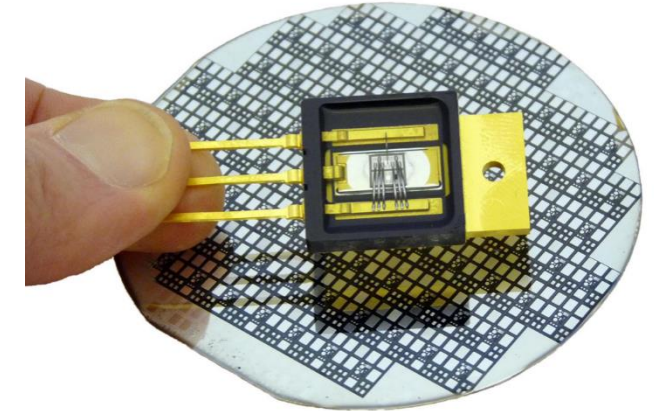
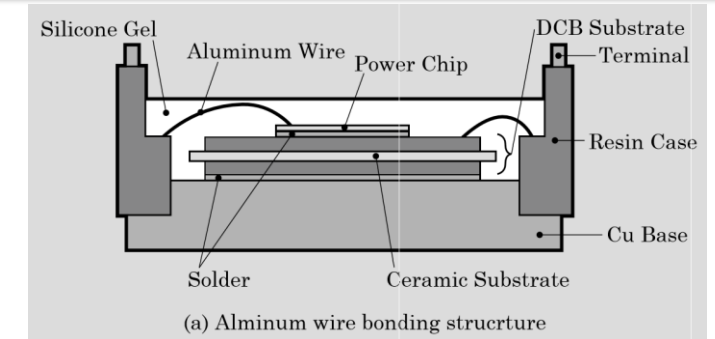


e.g:

- ☐ MOS transport studies
- ☐ Study traps in SiO₂ which are not accessible in Si devices
- ☐ Band-offsets and transport in oxide
- ☐ Interface traps and passivation
- ☐ Interface chemistry correlations

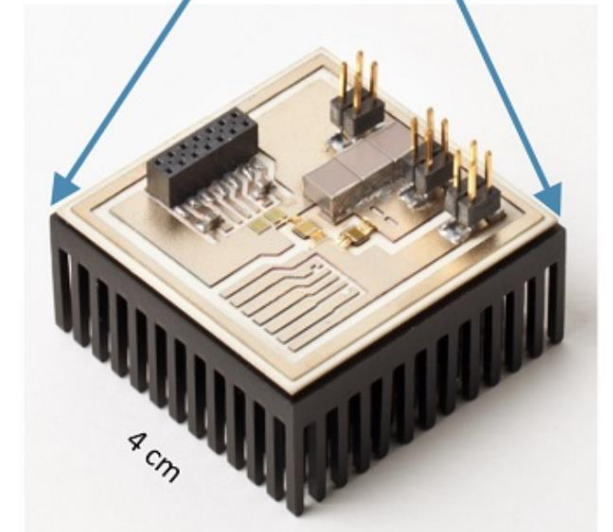
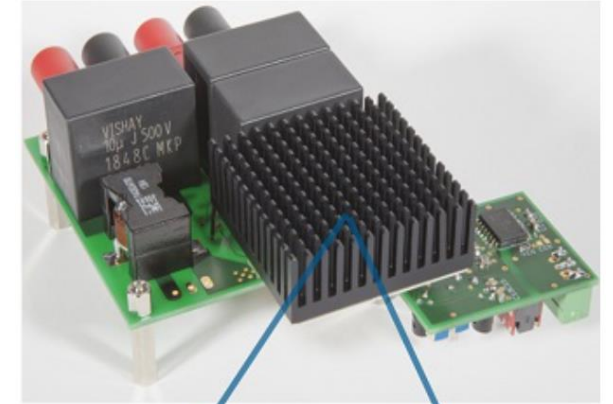
Integration, Packaging and Assembly issues

- High power density and high switching frequencies impose high requirements on packaging technologies. By appropriate approaches such as 3D packaging, parasitic effects that are caused by packages, substrates and layouts, can be systematically influenced and reduced. Due to compact hot spots, heat spreading and heat dissipation are more important than ever.



Efficiency, Power Density and Robustness

- Increasing the electrical efficiency of power electronic systems enables e.g. increased ranges in electromobility or higher profits in renewable energy conversion. One approach to reduce losses is the usage of **novel semiconductor materials** (e.g. GaN, SiC). These materials have superior physical properties compared to silicon-based power semiconductors. Another approach is to increase the **slew-rates** in power converters to reduce switching losses.
- Increasing the power density results in smaller passive components, smaller system components and thereby in potential weight, size and cost advantages. Approaches to increase the power density are the hybrid and monolithic integration of gate-driver and power converter circuits, as well as active and passive components.
- To ensure robustness of the systems, the thermal behavior and EMC of power modules is studied and optimized.

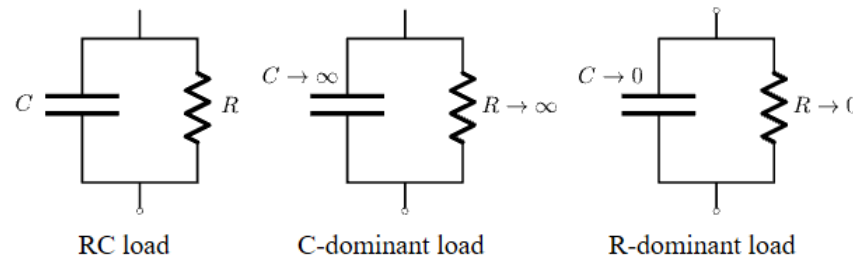


Gallium nitride based quasi-normally-off gate driver and boost converter

Recall: What is slew-rate

What is Slew-Rate (SR)?

Slew rate is the maximum voltage change per unit time in a node of a circuit, due to limited current sink or source. The SR of a circuit is limited by its slowest node, i.e., the one with the smallest Slew rate. Imagine a node in a circuit where the load is capacitive-dominant, i.e., the load is mostly made of a capacitance C (although there is always some very large parallel resistance R).



For simplicity let's assume the parallel resistance is infinite, so that all current I entering the load will flow to the capacitor and none to the resistance. Then, the rate of voltage change across the capacitor is defined by the well-known formula:

$$\frac{dV}{dt} = \frac{I}{C}.$$

If the current being sourced to this load is limited (and in reality always is), say at I_{max} , the maximum voltage change per unit time is:

$$SR = \max \left(\frac{dV}{dt} \right) = \frac{I_{max}}{C}$$

This is the Slew-Rate. If the current is being drawn from the capacitor, i.e., the capacitor is discharging, the same rule applies, except the voltage change is negative (negative SR).



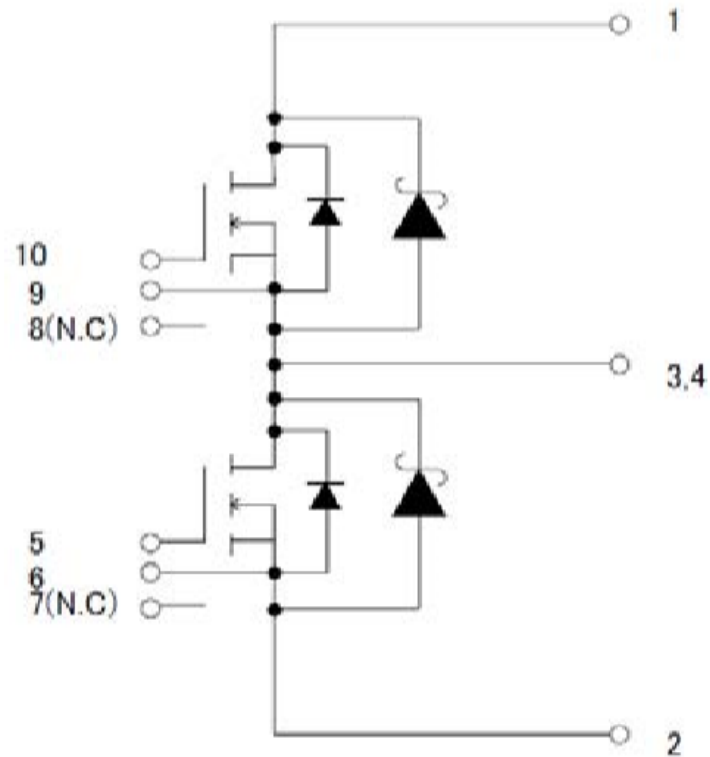
Failure analysis of SiC power devices

- **Gate threshold voltage degradation:** mainly due to the small thickness of gate oxide layer and the relatively high electric field applied. **Solution: optimize gate oxidation process**
- **Body diode degradation:** the body diode degrades under continuous forward current, resulting in higher forward voltages and ultimately higher power dissipation during conduction at a given current. **Solution: add a SiC Schottky Barrier Diode (SBD)**
- **Increased gate leakage current – a sign of device failure:** SiC MOSFET works at high frequency switching mode. The peak current will be 10 times higher than normal condition, causing rapid temperature rise and high leakage current. **Solution: add gate current monitoring system**
- **Avalanche events:** the voltage spike exceeds the breakdown voltage of the device, which then allows current to flow through the MOSFET, dissipating power and creating a sudden rise of the internal temperature. **Solution: more careful design and add extra cooling.**

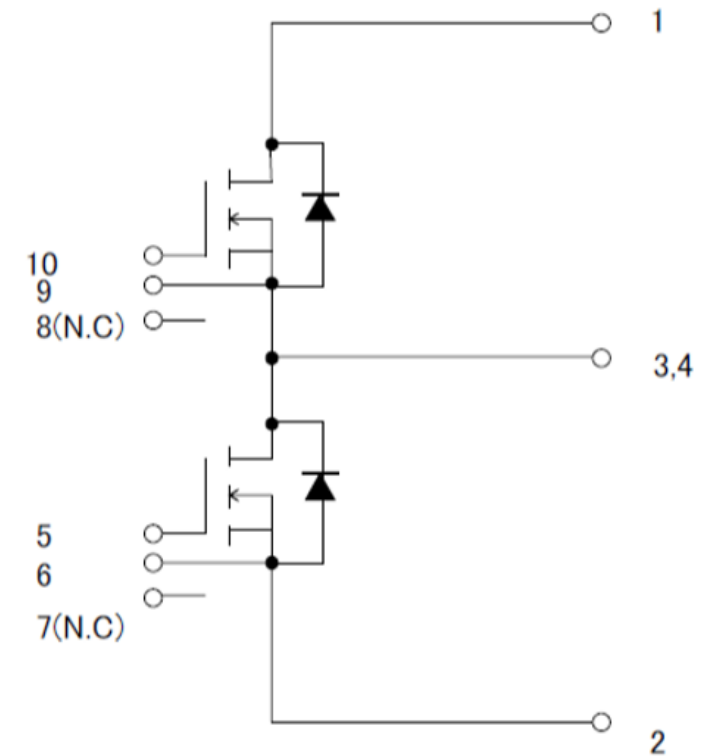


Effective circuits with/without SBD

■ SiC-MOSFET + SiC-SBD



■ SiC-MOSFET





What you can take away from this course

- Advantages and disadvantages of every SiC power devices
- Design considerations for a power switch
- Current status of the 3rd generation semiconductor research
- Mobility and on-resistance of SiC MOSFET
- Fabrication process
- Packaging and failure analysis

